

Measuring the performance of Recommender Systems

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O material e as imagens nestes slides são de (ou adaptadas de):

Charu C. Aggarwal, *Recommender Systems – The Textbook*, Springer, 2016 – Chapter 13

Accuracy metrics

1. prediction the accuracy of a **rating**
 2. prediction of the accuracy of **ranking** the items
- The ratings matrix is typically sampled in an entry-wise fashion
 - ❑ a subset of the entries are used for training
 - ❑ the remaining entries are used for accuracy evaluation (validation and testing)
 - ➔ this complicates things...
 - Different from standard classification and regression modelling that methods sample rows of the labelled data, rather than sampling the entries.
 - *Training/Validation/Test*

1. Accuracy at predicting rating values

- E is the set of entries in your test set
- $e_{ui} = \hat{r}_{ui} - r_{ui}$

$$MSE = \frac{\sum_{(u,j) \in E} e_{uj}^2}{|E|}$$

mean squared error

$$RMSE = \sqrt{\frac{\sum_{(u,j) \in E} e_{uj}^2}{|E|}}$$

root mean squared error

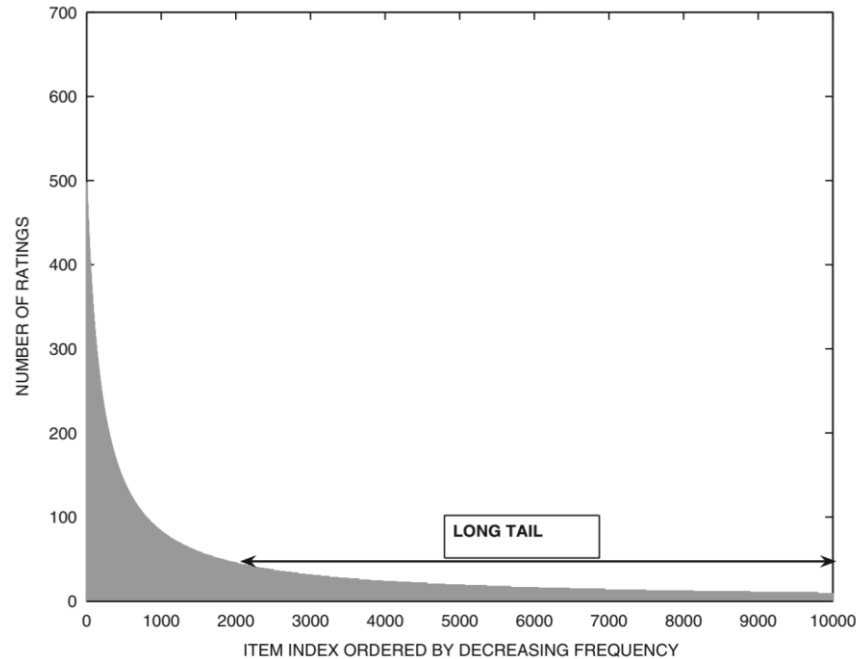
$$MAE = \frac{\sum_{(u,j) \in E} |e_{uj}|}{|E|}$$

mean-absolute-error

$$NRMSE = \frac{RMSE}{r_{max} - r_{min}}$$
$$NMAE = \frac{MAE}{r_{max} - r_{min}}$$

normalized RMSE &
normalized MAE

... remember the long tail...



- **items in the long tail often are the most important to predict**
- compute measures separately for all the hidden ratings associated with each item and then **average over the different items in a weighted way**.
- weight depends on the relative importance

2. Accuracy at predicting rankings

- the recommender system creates a ranking of items for a user, and the top-k items are recommended
- I_u set of items hold-out for user u

Question: how well the orderings of the ratings in I_u are related to the predicted orderings

- ratings are typically chosen from a discrete scale
- many ties exist in the ground truth (not penalize for rankings that are tied in the ground truth)

1) Spearman rank correlation coefficient: Pearson correlation coefficient applied on ranks.

- Can be averaged over all the users
- Can be calculated over all the ratings in the dataset

Can be *tie-corrected*.

2) Kendall rank correlation coefficient:

For each pair $i, k \in I_i$ calculate:

$$C(j, k) = \begin{cases} +1 & \text{if items } j \text{ and } k \text{ are in the same relative order in} \\ & \text{ground-truth ranking and predicted ranking (concordant)} \\ -1 & \text{if items } j \text{ and } k \text{ are in a different relative order in} \\ & \text{ground-truth ranking and predicted ranking (discordant)} \\ 0 & \text{if items } j \text{ and } k \text{ are tied in either the} \\ & \text{ground-truth ranking or predicted ranking} \end{cases}$$

$$\tau_u = \frac{\sum_{j < k} C(j, k)}{|I_u| \cdot (|I_u| - 1) / 2}$$

Kendall rank
correlation
coefficient for user u

$$\tau_u = \frac{\text{Number of concordant pairs} - \text{Number of discordant pairs}}{\text{Number of pairs in } I_u}$$

Can be averaged across all users.